Endpoint-24 Affine-Defect Boundary Atlas: B1/B2/B3, F_2^4 Error Geometry, and Finite Transport

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Abstract

We extract a paper-local finite theorem from the certified endpoint-24 order-four magic-square atlas. In value-minus-one coordinates over $\mathbb{F}_2^4$, the affine interpolation error $E = L + A$ has defect $\delta \in \{0, 4, 6\}$. On endpoint 24, this defect and the selected-label xor split the certified atlas into an exact affine core, a selected-affine B2 boundary, and a selected-non-affine B3 boundary:

$$236 = 144 + 32 + 60.$$

The verified free quotient $T/K \cong V_4$ gives

$$59 = 36 + 8 + 15.$$

The B3 support-action matrix has 64 candidate signature/profile cells, with exactly 15 realized and 49 absent cells. The eight B3 profile columns are now certified intrinsic to the selected signature and bilinear error matrix; the remaining realization cut is certified finite and integer-arithmetic, not an $\mathbb{F}_2$-linear closure. We separate the first-principles part of the argument, namely the degree/defect and error-fiber parity spine, from the finite transport and matrix certificates. The symbolic derivation of the B3 count 60 remains an explicit open debt.

Keywords. Magic squares, Dürer square, affine geometry over F_2, finite certificates, endpoint atlas.

1 Introduction

This paper is a focused companion to the public order-four magic-square work. Its object is narrower: the certified endpoint-24 boundary atlas in affine-defect coordinates. The earlier paper isolates 24 as a structured bounded meet. Here we examine the finite boundary that appears when the order-four endpoint-24 atlas is organized by affine error over $\mathbb{F}_2^4$.

The central split is

$$236 = 144 + 32 + 60,$$

where the three terms are the exact affine core, the selected-affine B2 boundary, and the selected-non-affine B3 boundary. The verified free $T/K \cong V_4$ action gives the quotient split

$$59 = 36 + 8 + 15.$$

The exact affine core has 144 records. The selected-affine boundary B2 has 32 records. The selected-non-affine boundary B3 has 60 records and carries a support-action matrix with 15 realized and 49 absent signature/profile cells. B1 is included as a scoped companion statement: the endpoint 23/27 signatures are absent from the affine selected-plane layer by the same four-point xor criterion, without claiming a full endpoint 23/27 classification.

Contribution. The paper has three layers.

1. It states the endpoint-24 affine-defect boundary theorem in a compact paper-local form.
2. It separates the structural part of the proof from the finite matrix part. The degree/defect dichotomy and the error-fiber parity bridge are proved as first-principles finite theorems; the B3 support-action matrix is a finite transport certificate.
3. It collapses the remaining B3 count debt from "why 15 orbits" to the sharper finite kernel

$$\delta = 0, 4, 6 \mapsto 36, 16, 7$$

and the B3 row-degree pattern $1, 1, 1, 2, 2, 1, 4, 3$.

Claim discipline. Replay-backed statements are labelled as Certified Propositions and point to the source map in `paper/SOURCE_MAP.md`. The manuscript does not claim a non-enumerative derivation of the B3 count 60, a symbolic derivation of the endpoint kernel $36, 16, 7$, a full endpoint $23/27$ classification, or a full classification of non-affine order-four magic-square behavior.

2 Model and Notation

Cells are indexed by $\mathbb{F}_2^4$, and square values are converted to labels by subtracting one. Thus the values $1, \ldots, 16$ become labels $0, \ldots, 15$. For a normal order-four magic square, let

$$L : \mathbb{F}_2^4 \to \mathbb{F}_2^4$$

be the label map. Let A be the affine interpolation of L on the five base points $\{0, e_1, e_2, e_3, e_4\}$, and define the error map

$$E = L + A, \qquad \delta = |\operatorname{supp}(E)|.$$

For four distinct selected labels, the selected set is an affine plane exactly when its xor is zero. This selected-xor axis is the finite selector used to separate the exact affine row, B2, and B3 on endpoint 24.

Definition 2.1 (Square-mask pair and endpoint)**.** A square-mask pair is a normal order-four magic square together with a one-incidence mask. The mask selects one cell in each row, one cell in each column, and one cell on each of the two main diagonals. The endpoint is $34 - t$, where $t = \min_{c \in M} L(c)$ is the minimum selected label (the minimum selected value minus one). Equivalently, endpoint 24 is exactly the slice whose minimum selected value is 11. This paper works inside the certified endpoint 24 subatlas, which contains 236 such records. In particular every endpoint-24 selected four-set contains the value 11, which is why the B3 signatures are four-subsets of $\{11, \ldots, 16\}$ containing 11.

Definition 2.2 (Selected mask and selected labels)**.** For a square-mask pair, $M \subset \mathbb{F}_2^4$ denotes the four selected cells. The selected labels are the four elements $L(M) \subset \mathbb{F}_2^4$. The selected set is affine if and only if

$$\bigoplus_{c \in M} L(c) = 0.$$

Equivalently, in value notation, the selected values form an affine four-set after subtracting one from each value.

Definition 2.3 (Endpoint-24 record)**.** An endpoint-24 record is a certified square-mask pair whose endpoint is 24. The development atlas contains 236 such records.

Definition 2.4 (Cell-value affineness and essential representatives)**.** A record is cell-value affine when its full label map $L : \mathbb{F}_2^4 \to \mathbb{F}_2^4$ is affine. The finite atlas is stored modulo the usual dihedral D_4 symmetries of the square; theorem statements that count quotient objects explicitly say when the free T/K quotient is also being used.

Definition 2.5 (The T/K quotient)**.** The finite transport group T is the row/column relation group used by the extension certificates, and K is its loop stabilizer. The verified quotient $T/K \cong V_4$ acts freely on every certified square-mask pair and preserves the endpoint, affine-defect class, and selected signature. Hence each T/K-orbit has four records.

Definition 2.6 (Signature and support-action profile)**.** For B3, a signature is one of the eight non-affine selected-label four-sets containing the minimum selected value 11. A support-action profile records the defect δ, the error-direction space $\mathcal{D}$, and the finite defect-mask support type seen by the selected mask. The B3 support-action matrix has these eight signatures as rows and the eight realized profiles as columns.

Definition 2.7 (Clause templates)**.** The public names NF1, NF2, and NF3 refer to the three B3 support-action clause templates. NF1 is the defect-4 singleton xor-error case. NF2 is the defect-6 non-xor pair-support case. NF3 is the defect-6 non-xor triple-support case. The templates are structural; the thirteen profile-level instances are finite-certified.

Definition 2.8 (Affine defect)**.** The affine defect of a record is $\delta = |\operatorname{supp}(E)|$, where $E = L + A$. The associated error direction space $\mathcal{D}$ is the span of the bilinear error coefficients. The certified dichotomy is

$$(\delta, \dim \mathcal{D}) \in \{(0,0), (4,1), (6,2)\}.$$

Definition 2.9 (Boundary labels)**.** Inside endpoint 24, we use the following names.

Name	Selector	Records	T/K-orbits
exact affine core	selected affine, $\delta = 0$	144	36
B2	selected affine, $\delta = 4$	32	8
B3	selected non-affine, $\delta \in \{4, 6\}$	60	15

The classical context is the order-four magic-square structure of Ollerenshaw and Bondi [3], vector-space viewpoints on magic squares [6], the Dürer tesseract interpretation [4], and enumeration work on magic squares and the order-four case [1, 2]. A recent computational classification of the 880 essentially different normal fourth-order magic squares is recorded by Takemura [5]. The operative finite proofs in this paper are the local certificates listed in Section 5.

3 Algebraic Spine and Boundary Theorem

The theorem below states the paper-local boundary result in public notation. Each finite assertion is tied to a source artifact in `paper/SOURCE_MAP.md`; the proof uses those artifacts as certified finite inputs rather than as hidden mathematical assumptions.

Certified Proposition 3.1 (Line-xor and degree-bound certificate)**.** *In value-minus-one labels, every row and every column of each certified normal order-four magic square has xor zero. This is the local Nimm0/Lemma B input. The manuscript treats the Step4 finite derivation as the proof source; the classical order-four literature supplies context rather than an unverified quoted theorem [3, 6]. With the row/column line-xor input, the degree-bound argument gives*

$$\deg L \leq 2.$$

Proof. Write each output bit of $L(r_1, r_0, c_1, c_0)$ in algebraic normal form over $\mathbb{F}_2$. For a monomial $r^{S_r}c^{S_c}$, xoring over a fixed row leaves it alive exactly when $S_c = \{c_1, c_0\}$. Hence the condition that every row xors to zero kills precisely the monomials divisible by c_1c_0. Similarly, column-xor zero kills precisely the monomials divisible by r_1r_0. Every degree-three or degree-four monomial in the four variables is divisible by one of these two pairs, so no monomial of degree at least three remains. The row/column line-xor premise itself is the Step4 finite row-pair derivation from the full row, column, and diagonal magic constraints. □

Lemma 3.2 (Quadratic error form)**.** *Let A be the affine interpolation of L on $\{0, e_1, e_2, e_3, e_4\}$, and set $E = L + A$. For the certified normal magic-square records, the pure row-pair and pure column-pair quadratic coefficients vanish. Hence the error has bilinear form*

$$E(r, c) = B_{11}r_1c_1 + B_{10}r_1c_0 + B_{01}r_0c_1 + B_{00}r_0c_0.$$

Proof. The affine interpolation A absorbs the constant and linear parts of L. By Proposition 3.1, no terms of degree at least three occur. The row-xor condition kills the pure column-pair term c_1c_0, while the column-xor condition kills the pure row-pair term r_1r_0. The only quadratic terms left are the four mixed row/column monomials displayed above. □

Theorem 3.3 (Defect dichotomy / Lemma C)**.** *For the degree-at-most-two bijections of $\mathbb{F}_2^4$ covered by the certificate, the bilinear coefficient matrix has the certified one-dependent-side property. Consequently the error-direction span $\mathcal{D}$ has dimension at most two and*

$$(\delta, \dim \mathcal{D}) \in \{(0, 0), (4, 1), (6, 2)\}.$$

This is the first-principles structural part of the affine-defect spine.

Proof. Let $M = (B_{ij})$ be the 2×2 matrix of mixed bilinear coefficients. The property that a bilinear matrix is completable to a degree-at-most-two bijection $x \mapsto A_0x + E(x)$, and the property that one side of M is dependent, are invariant under the codomain action of $GL(4, 2)$. The Step5+ check therefore reduces the obstruction to a symmetry-complete finite set: all 4096 matrices with entries in a fixed three-dimensional subspace, plus one representative for the four-dimensional span case. In that reduced universe every completable matrix has one dependent side, and the four-dimensional representative is not completable. Thus a quadratic bijection has $\dim \mathcal{D} \le 2$ and a dependent side.

With one dependent side, E factors as a nonzero scalar linear form on one $\mathbb{F}_2^2$ coordinate side times a vector-valued linear map on the other side. If $d = \dim \mathcal{D}$, the scalar form is nonzero on two points and the vector map is nonzero on $4 - 2^{2-d}$ points. Hence

$$\delta = 2(4 - 2^{2-d}),$$

which gives exactly $(0, 0), (4, 1), (6, 2)$. □

Lemma 3.4 (Error-fiber parity bridge)**.** *Let M be the selected mask. If $\delta = 4$, then $\mathcal{D} = \{0, u\}$ and $E = u$ on a four-cell support Σ, so*

$$\bigoplus_{c \in M} L(c) = u \cdot (|M \cap \Sigma| \bmod 2).$$

If $\delta = 6$, the three nonzero fibers $\Sigma_u, \Sigma_v, \Sigma_{u+v}$ contribute

$$\bigoplus_{c \in M} L(c) = (|M \cap \Sigma_u| \bmod 2)u + (|M \cap \Sigma_v| \bmod 2)v + (|M \cap \Sigma_{u+v}| \bmod 2)(u + v).$$

Thus selected-label affineness is controlled by error-fiber parity.

Proof. The selected masks in this endpoint atlas are affine planes in the cell space, so the affine part cancels:

$$\bigoplus_{c\in M} A(c) = 0.$$

Thus $\bigoplus_{c\in M} L(c) = \bigoplus_{c\in M} E(c)$. The factorization in Theorem 3.3 gives the fiber shapes: for $\delta = 4$ a single nonzero error value u occurs on a four-cell support, and for $\delta = 6$ the three nonzero values $u, v, u+v$ occur on three two-cell fibers. Grouping the xor of E over M by these fibers gives exactly the two displayed parity formulas. □

Certified Proposition 3.5 (Free T/K-transport)**.** *The finite transport certificate proves a free global $T/K \cong V_4$ action on the certified atlas. It preserves endpoint, class, defect, and selected signature. Therefore endpoint* 24 *has* $236 = 59 \cdot 4$*, with quotient split*

$$59 = 36 + 8 + 15.$$

Certified Proposition 3.6 (B2 selected-affine boundary)**.** *Inside endpoint* 24*, B2 is the selected-affine, cell-value-non-affine boundary. It has* 32 *records and* 8 *free T/K-orbits. In the $\delta = 4$ fiber picture it is the even side of the constant-error support parity:*

$$|M \cap \Sigma| \equiv 0 \pmod 2.$$

The endpoint package verifies this selected-affine, cell-value-non-affine finite selector and closes the 32/8 *count inside the saved endpoint atlas. This is a structural parity selector plus a finite endpoint package, not a complete conceptual B2 classification theorem.*

Certified Proposition 3.7 (B3 support-action matrix)**.** *Inside endpoint* 24*, B3 is the selected-non-affine boundary. It has* 60 *records and* 15 *free T/K-orbits. The support-action matrix has eight non-affine selected signatures against eight support-action profiles, hence* 64 *candidate cells. The finite bilinear transport package verifies exactly* 15 *realized cells and* 49 *absent cells, with no false positives or false negatives. The intrinsic column-classification certificate refines the column side: all eight profile columns are intrinsic to the selected signature and the bilinear error matrix M, through δ, the zero pattern of M, and $\mathcal{D} = \operatorname{span}(M)$. The defect-6 realizability-obstruction certificate identifies the remaining realization side as an integer-magic residue: at defect* 6*, the order-four magic-square constraints realize* 13 *of the* 35 *two-dimensional error spaces, although all* 35 *are $\mathbb{F}_2$-completable and the realized* 13*-set has trivial $GL(4,2)$ stabilizer. Thus the column classification is intrinsic, while the* 15/49 *realization cut remains a finite arithmetic certificate. The NF1/NF2/NF3 templates are structural clause shapes; the thirteen profile-level instances are finite-certified.*

Certified Proposition 3.8 (B3 count collapse)**.** *The quotient count* 15 *is not an independent count once freeness is known:*

$$15 = 60/4.$$

The finite residual is the row-degree pattern

$$1, 1, 1, 2, 2, 1, 4, 3$$

and the endpoint quotient kernel

$$\delta = 0, 4, 6 \mapsto 36, 16, 7.$$

This is a finite debt reduction, not a symbolic proof of the count 60*. A non-enumerative derivation of that count remains outside the claim.*

Certified Proposition 3.9 (B1 scoped affine-plane absence)**.** *B1 is the scoped endpoint* $23/27$ *absence statement for the affine selected-plane layer. In that layer every selected four-set has selected-label xor zero. The endpoint* $23/27$ *signatures under discussion have nonzero selected-label xor, and therefore cannot occur in the affine selected-plane layer. This is not a full endpoint* $23/27$ *classification.*

Theorem 3.10 (Affine-defect boundary atlas, paper-local form)**.** *Inside the certified endpoint-24 order-four magic-square atlas, consider square-mask pairs in the bounded one-incidence model. Quotient statements are taken modulo the verified* D_4 *essential representatives and the verified free* T/K *action. Then:*

1. *endpoint* 24 *has* 236 *records and* 59 *free* T/K*-orbits;*
2. *the affine defect satisfies* $\delta \in \{0, 4, 6\}$*, with record split* $144/64/28$*;*
3. *the exact affine core has* 144 *records and* 36 T/K*-orbits;*
4. *B2 has* 32 *records and* 8 T/K*-orbits;*
5. *B3 has* 60 *records and* 15 T/K*-orbits. Its support-action matrix has* 64 *candidate cells, exactly* 15 *realized cells, and exactly* 49 *absent cells;*
6. *B1 is the scoped endpoint* $23/27$ *absence statement from the affine selected-plane layer, not a full endpoint* $23/27$ *classification.*

Proof. Proposition 3.1 gives $\deg L \leq 2$ from row/column line xor zero, and Lemma 3.2 writes $E = L + A$ as a bilinear error on row/column coordinates. Theorem 3.3 gives $\delta \in \{0, 4, 6\}$ with $(\delta, \dim \mathcal{D}) = (0, 0), (4, 1), (6, 2)$, and the error-fiber parity bridge identifies the selected-label xor from the selected mask's intersections with the error fibers.

The free T/K-transport proposition gives $236 = 59 \cdot 4$ on endpoint 24. The exact affine core is the selected-affine, $\delta = 0$ cell and has 144 records, hence 36 free T/K-orbits. The B2 proposition gives the selected-affine boundary as 32 records and 8 orbits. The B3 support-action proposition gives the selected-non-affine boundary as 60 records and 15 orbits, with the $15/49$ support-action matrix. The count-collapse proposition records the sharper finite residual without turning it into a symbolic count proof. Finally, the B1 proposition gives the scoped endpoint 23/27 absence from the affine selected-plane layer. These blocks are exactly the assertions of the theorem. □

4 Source-Backed Tables and Figures

The following compact displays are generated from the certified JSON artifacts listed in `paper/SOURCE_MAP.md`. They expose only paper-facing aggregate data, not record-level payloads.

$$236 = 144 + 32 + 60, \qquad 59 = 36 + 8 + 15.$$

Figure 1: Endpoint-24 record and quotient splits. Source-map rows: endpoint split, free transport, and count collapse.

Selected affine	Defect δ	Records	Class
yes	0	144	exact affine
yes	4	32	B2
yes	6	0	empty
no	0	0	empty
no	4	32	B3
no	6	28	B3

Table 1: Endpoint-24 two-invariant table: selected-label affineness against affine defect.

δ	Records	T/K-orbits	Preserved diagonal planes
0	144	36	24
4	64	16	8
6	28	7	0

Table 2: Endpoint quotient kernel and finite diagonal-plane avatar. The last column is certified finite data, not a first-principles derivation.

Signature	xor	records	orbits	degree	δ split
11,12,13,15	3	4	1	1	d4:4
11,12,13,16	2	4	1	1	d4:4
11,12,14,15	2	4	1	1	d4:4
11,12,14,16	3	8	2	2	d4:4, d6:4
11,13,14,15	5	8	2	2	d4:4, d6:4
11,13,14,16	4	4	1	1	d4:4
11,13,15,16	7	16	4	4	d6:16
11,14,15,16	6	12	3	3	d4:8, d6:4

Table 3: B3 selected-signature degrees. The degree column is the row sum in the support-action matrix.

Signature	DM01	DM02	DM03	DM04	DM05	DM06	DM07	DM08	degree
11,12,13,15	1	.	.	.	.	.	.	.	1
11,12,13,16	.	1	.	.	.	.	.	.	1
11,12,14,15	.	1	.	.	.	.	.	.	1
11,12,14,16	1	.	1	.	.	.	.	.	2
11,13,14,15	1	.	.	.	.	.	1	.	2
11,13,14,16	.	1	.	.	.	.	.	.	1
11,13,15,16	.	.	.	1	1	1	.	1	4
11,14,15,16	1	1	.	.	.	.	1	.	3

Table 4: B3 support-action matrix. Entry 1 means realized; a dot means absent.

Profile	δ	orbits	allowed xors	families
DM01	4	4	3,5,6	NF1
DM02	4	4	2,4,6	NF1
DM03	6	1	3	NF3
DM04	6	1	7	NF2
DM05	6	1	7	NF2
DM06	6	1	7	NF2
DM07	6	2	5,6	NF3
DM08	6	1	7	NF3

Table 5: Eight B3 support-action profiles, with allowed selected-label xors and clause families.

Certified block	Certified content	Paper consequence
Intrinsic columns	8/8 B3 profile columns intrinsic to (S,M)	the column side has no mask-transport residue
Defect-6 residue	defect 6 realizes 13 of 35 two-flats; all 35 are F2-completable	the realization cut is integer-magic, not F2-linear
Defect-6 residue	the realized 13-set has GL(4,2) stabilizer order 1	no linear symmetry selects the remaining cut

Table 6: Post-obstruction B3 status. The intrinsic-column certificate closes the profile-column side; the defect-6 obstruction identifies the remaining realization cut as integer-magic rather than $\mathbb{F}_2$-linear.

Family	Clauses	Public rule
NF1	6	defect 4; singleton xor-error direction
NF2	3	defect 6; non-xor pair witness
NF3	4	defect 6; non-xor triple witness

Table 7: NF1/NF2/NF3 clause-template counts.

Clause	Family	Profile	x	E	witness
`L20-C01`	NF1	`DM01`	3	`3`	`-`
`L20-C02`	NF1	`DM01`	5	`5`	`-`
`L20-C03`	NF1	`DM01`	6	`6`	`-`
`L20-C04`	NF1	`DM02`	2	`2`	`-`
`L20-C05`	NF1	`DM02`	4	`4`	`-`
`L20-C06`	NF1	`DM02`	6	`6`	`-`
`L20-C07`	NF3	`DM03`	3	`3,13,14`	`14`
`L20-C08`	NF2	`DM04`	7	`2,5,7`	`2,5`
`L20-C09`	NF2	`DM05`	7	`1,6,7`	`1,6`
`L20-C10`	NF2	`DM06`	7	`3,4,7`	`3,4`
`L20-C11`	NF3	`DM07`	5	`5,8,13`	`8`
`L20-C12`	NF3	`DM07`	6	`6,8,14`	`8`
`L20-C13`	NF3	`DM08`	7	`7,11,12`	`11`

Table 8: The thirteen finite profile-level clause instances. Empty witnesses are shown as `-`.

5 Certificate Spine

The paper-facing certificate table is maintained in `paper/SOURCE_MAP.md`. Before public export, every displayed number in the manuscript must be tied to one source artifact and one regression test. The local test count is deliberately not hardcoded in the mathematical text; it is a build artifact recorded by the current regression run.

The key public-safe statement is that the B3 matrix is finite-certified, while the symbolic derivation of the count 60 remains open. The count-collapse certificate reduces that debt to the finite endpoint quotient kernel

$$\delta = 0, 4, 6 \mapsto 36, 16, 7$$

and to the B3 row-degree pattern

$$1, 1, 1, 2, 2, 1, 4, 3.$$

The intrinsic column-classification certificate closes the column side by proving that all eight B3 profile columns are intrinsic to (S, M). The defect-6 realizability-obstruction certificate does not close the realization side; it proves that the defect-6 residue is an irreducible integer-magic cut, not a consequence of $\mathbb{F}_2$-completability or linear symmetry.

Certified Proposition 5.1 (Reproducibility artifact table)**.** *The manuscript depends on the following certified artifacts. Exact file names are kept in `paper/SOURCE_MAP.md`; this table uses only reproducibility handles and mathematical block names.*

Claim block	*Reproducibility handle*	*Status*
endpoint split	*endpoint split certificate*	*certified*
defect dichotomy	*first-principles defect dichotomy*	*structural*
error parity	*error-fiber parity bridge*	*structural*
B3 matrix	*support-action matrix summary*	*finite*
B3 intrinsic columns	*intrinsic column classification*	*finite*
B3 defect-6 residue	*defect-6 realizability obstruction*	*finite*
endpoint transport	*endpoint transport package*	*finite*
T/K freeness	*global free-transport theorem*	*finite*
B2 package	*B2 endpoint boundary package*	*finite*
count collapse	*count-collapse kernel geometry*	*finite*

Remark 5.2 (Attribution and proof source). The row/column nim-sum property is positioned against the classical order-four literature, including Ollerenshaw–Bondi and Ward. Until a precise external bibliographic statement is pinned down, the manuscript treats the local Step4 derivation as the proof source.

6 Public Companion Package

The public companion repository follows the curated package structure

`paper/`, `data/`, `docs/`, `scripts/`, `tests/`, `results/`.

It is a curated replay and audit package, not a dump of the development workspace. Public artifacts are limited to the claims actually made in the manuscript, and the blocked claims in `paper/CLAIM_BOUNDARY.md` remain blocked.

Blocked claims. The public package does not claim public record-level widening, full endpoint 23/27 classification, complete conceptual B2 theory, full B3 non-affine classification, a symbolic derivation of the B3 15/49 matrix, or a non-enumerative derivation of the count 60.

Release artifacts. The package includes curated JSON summaries, source-map and claim-boundary documents, audit scripts, pytest smoke tests, `README.md`, `REPRODUCE.md`, `CITATION.cff`, and `LICENSE`. Its replay scope is intentionally narrower than the development workspace and traces to `paper/SOURCE_MAP.md`.

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